

# Model Evaluation

## ADS 503 Final Team Project - Diabetes Health Indicators (BRFSS 2015)

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### Table of contents

#### Libraries

```
library(tidyverse)
library(caret)
library(recipes)
library(pROC)
library(scales)
library(doParallel)
library(glmnet)
library(ggplot2)
library(ranger)
library(gbm)
library(PRRROC) # Explicitly loaded for Precision-Recall evaluation metrics

set.seed(503)
dm_palette <- c("No Diabetes" = "#4C9F70", "Diabetes" = "#D1495B")
```

#### Importing from Model Tuning

```
setwd(r"(C:\Users\User\ADS503_Team5\Raw Data)")

load("tuned_models_and_data.RData")
#source: https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/load
```

#### ROC Curve Construction

```

#Source: Robin (2026). Package 'pROC'.
#Link: https://xrobin.github.io/pROC/
prob_lda <- predict(mod_glm, x_test_processed, type = "prob")[["Diabetes"]] #gemini suggested
prob_enet <- predict(mod_glmnet, x_test_processed, type = "prob")[["Diabetes"]] #gemini sugg
prob_rf <- predict(mod_rf, x_test_df, type = "prob")[["Diabetes"]]
#gemini suggested [["Diabetes"]] subset when calling predict()
prob_gbm <- predict(mod_gbm, x_test_df, type = "prob")[["Diabetes"]]
#gemini suggested [["Diabetes"]] subset when calling predict()

roc_lda <- roc(test_y, prob_lda, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_enet <- roc(test_y, prob_enet, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_rf <- roc(test_y, prob_rf, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
roc_gbm <- roc(test_y, prob_gbm, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)

```

DeLong Test Results for model ROC Curves

```

# Execute statistical comparison of holdout curves
#LDA Vs. Elastic Net
delong_result <- roc.test(roc_lda, roc_enet, method = "delong")
print(delong_result)

```

DeLong's test for two correlated ROC curves

```

data:  roc_lda and roc_enet
Z = -0.45462, p-value = 0.6494
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.0001649040  0.0001028077
sample estimates:
AUC of roc1 AUC of roc2
 0.8272879  0.8273189

```

```

#Random Forest vs. GBM
delong_result2 <- roc.test(roc_rf, roc_gbm, method = "delong")
print(delong_result2)

```

DeLong's test for two correlated ROC curves

```
data:  roc_rf and roc_gbm
Z = -1.6704, p-value = 0.09484
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.0031871807  0.0002542256
sample estimates:
AUC of roc1 AUC of roc2
 0.8287006   0.8301671
```

```
#LDA versus GBM
delong_result3 <- roc.test(roc_lda, roc_gbm,method = "delong")
print(delong_result3)
```

DeLong's test for two correlated ROC curves

```
data:  roc_lda and roc_gbm
Z = -3.0315, p-value = 0.002433
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.004740708 -0.001017740
sample estimates:
AUC of roc1 AUC of roc2
 0.8272879   0.8301671
```

```
#Random Forest versus LDA
delong_result4 <- roc.test(roc_rf, roc_lda,method = "delong")
print(delong_result4)
```

DeLong's test for two correlated ROC curves

```
data:  roc_rf and roc_lda
Z = 1.1642, p-value = 0.2443
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.000965619  0.003791111
sample estimates:
AUC of roc1 AUC of roc2
 0.8287006   0.8272879
```

```
#Random Forest versus Elastic Net
delong_result5 <- roc.test(roc_rf, roc_enet,method = "delong")
print(delong_result5)
```

DeLong's test for two correlated ROC curves

```
data:  roc_rf and roc_enet
Z = 1.1538, p-value = 0.2486
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 -0.0009653919  0.0037287879
sample estimates:
AUC of roc1 AUC of roc2
 0.8287006  0.8273189
```

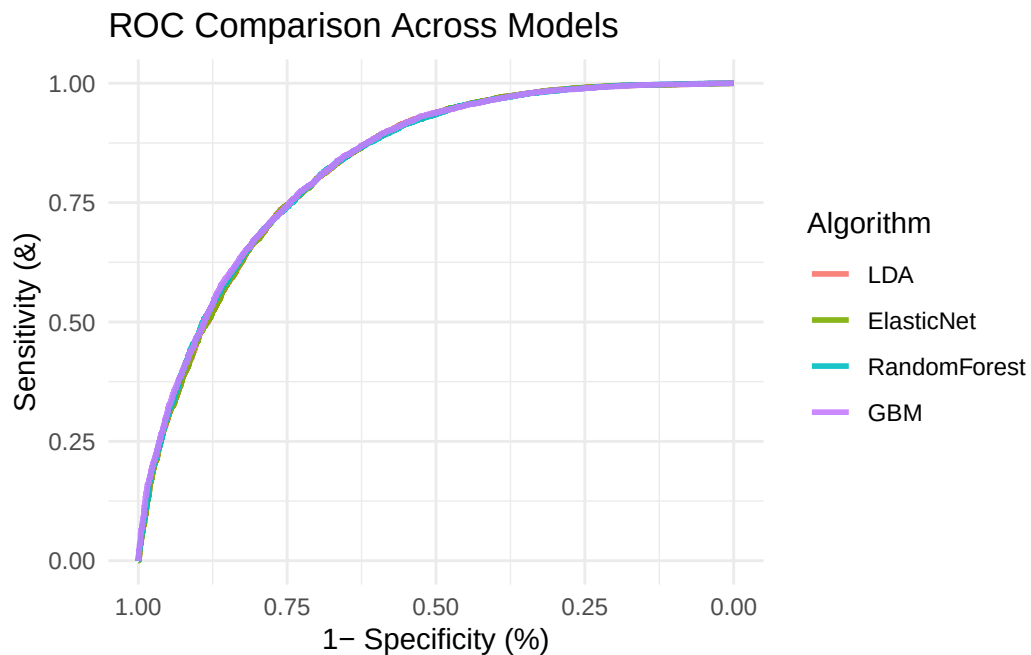
Judging by the results of the DeLong's tests, most of the models remain statistically indistinguishable from each other except for LDA and GBM, where the p value of that test is 0.002433. GBM outperforms LDA significantly.

ROC Comparison

```
##Source: Robin (2026). Package 'pROC'.
#Link:  https://xrobin.github.io/pROC/
# Build individual ROC objects first
r_lda   <- roc(test_y, prob_lda,  levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_enet  <- roc(test_y, prob_enet, levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_rf    <- roc(test_y, prob_rf,   levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)
r_gbm   <- roc(test_y, prob_gbm,  levels = c("NoDiabetes", "Diabetes"), quiet = TRUE)

# Put them in a named list
#suggested by Gemini:
roc_list <- list(LDA = r_lda, ElasticNet = r_enet, RandomForest = r_rf, GBM = r_gbm)

# Plot all curves simultaneously with ggplot2
#source : https://www.rdocumentation.org/packages/ggROC/versions/1.0/topics/ggroc
ggroc(roc_list, aes = "color", linewidth = 1, alpha = 0.9) +
  theme_minimal() +
  labs(title = "ROC Comparison Across Models", color = "Algorithm", x = "1- Specificity (%)")
```



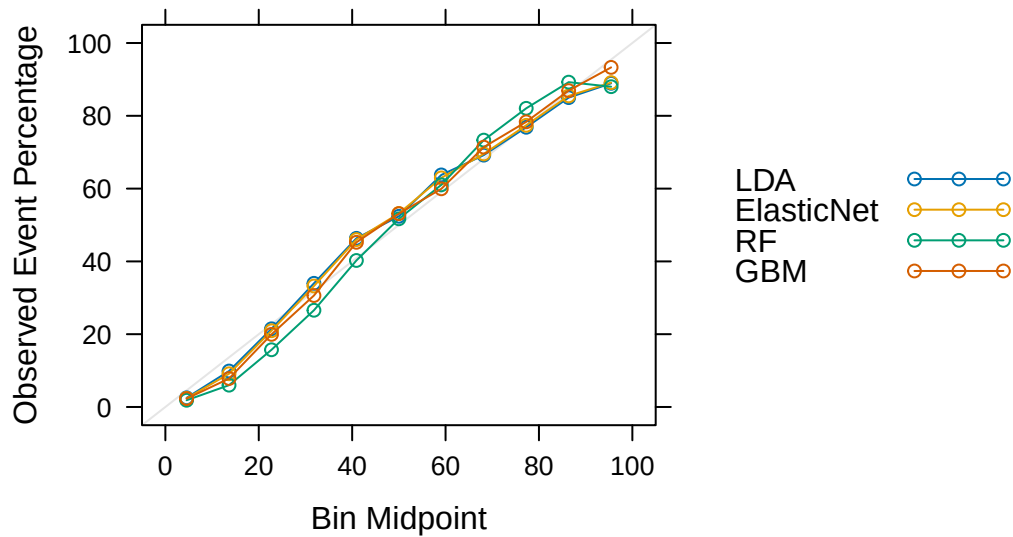
### Probability Calibration Assessment

```
# Create a dedicated dataframe for calibration
calibration_df <- data.frame(
  obs      = test_y,
  LDA      = prob_lda,
  ElasticNet = prob_enet,
  RF       = prob_rf,
  GBM      = prob_gbm
) #suggested by Gemini

#source: Kuhn (2026). Package 'caret': 'calibration'.
#https://cran.r-project.org/web/packages/caret/caret.pdf

#source for auto.key
#https://stackoverflow.com/questions/10828016/lattice-auto-key-how-to-adjust-lines-and-points
cal_obj <- calibration(obs ~ LDA+ElasticNet+RF+GBM, data = calibration_df, cuts = 11)
plot(cal_obj, type = "b", main = "Probability Calibration Curve", auto.key = TRUE)
```

## Probability Calibration Curve



Model summaries

```
#Creating a list of the models loaded in
#source: https://www.w3schools.com/r/r\_lists.asp

models <- list(
  LDA      = mod_glm,
  ElasticNet = mod_glmnet,
  RandomForest = mod_rf,
  GBM      = mod_gbm
)

#Using resamples to give a summary of the models
#https://www.rdocumentation.org/packages/caret/versions/6.0-92/topics/resamples
res <- resamples(models)
summary(res)
```

Call:

```
summary.resamples(object = res)
```

Models: LDA, ElasticNet, RandomForest, GBM

Number of resamples: 5

ROC

|              | Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      | NA's |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|------|
| LDA          | 0.8196668 | 0.8212530 | 0.8237375 | 0.8240489 | 0.8259243 | 0.8296629 | 0    |
| ElasticNet   | 0.8197351 | 0.8213236 | 0.8237687 | 0.8240887 | 0.8258696 | 0.8297466 | 0    |
| RandomForest | 0.8213277 | 0.8240323 | 0.8240529 | 0.8259629 | 0.8281790 | 0.8322225 | 0    |
| GBM          | 0.8216358 | 0.8252026 | 0.8255894 | 0.8265866 | 0.8293572 | 0.8311482 | 0    |

#### Sens

|              | Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      | NA's |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|------|
| LDA          | 0.7598585 | 0.7671499 | 0.7675035 | 0.7677618 | 0.7685234 | 0.7757737 | 0    |
| ElasticNet   | 0.7584439 | 0.7664427 | 0.7669731 | 0.7673728 | 0.7697613 | 0.7752431 | 0    |
| RandomForest | 0.7842617 | 0.7854996 | 0.7896040 | 0.7905364 | 0.7929632 | 0.8003537 | 0    |
| GBM          | 0.7709991 | 0.7736516 | 0.7756365 | 0.7761432 | 0.7766973 | 0.7837312 | 0    |

#### Spec

|              | Min.      | 1st Qu.   | Median    | Mean      | 3rd Qu.   | Max.      | NA's |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|------|
| LDA          | 0.7253758 | 0.7282051 | 0.7282532 | 0.7286486 | 0.7299735 | 0.7314356 | 0    |
| ElasticNet   | 0.7259063 | 0.7282051 | 0.7285588 | 0.7284364 | 0.7287836 | 0.7307284 | 0    |
| RandomForest | 0.7024399 | 0.7082228 | 0.7122900 | 0.7101179 | 0.7137553 | 0.7138815 | 0    |
| GBM          | 0.7161804 | 0.7197171 | 0.7238331 | 0.7230257 | 0.7256011 | 0.7297966 | 0    |

Highest Mean ROC: GBM Highest Mean Sensitivity: Random Forest Highest Mean Specificity: LDA

#### Model Comparison

```
bestTunes <- list(
  LDA      = mod_glm$bestTune,
  ElasticNet = mod_glmnet$bestTune,
  RandomForest = mod_rf$bestTune,
  GBM      = mod_gbm$bestTune
)
print(bestTunes)
```

#### \$LDA

```
parameter
1      none
```

#### \$ElasticNet

```
alpha      lambda
27 0.4 0.002674312
```

#### \$RandomForest

```

      mtry splitrule min.node.size
1      2      gini              1

$GBM
      n.trees interaction.depth shrinkage n.minobsinnode
1      150              1      0.1              50

#Using diff to give a summary of the resamples
#https://www.rdocumentation.org/packages/caret/versions/6.0-92/topics/resamples
resamples_diff <- diff(res) #gemini suggested using diff() on the entire res (resample) vari
summary(resamples_diff) #gemini suggested the use of summary() for the final results

```

Call:

```
summary.diff.resamples(object = resamples_diff)
```

p-value adjustment: bonferroni

Upper diagonal: estimates of the difference

Lower diagonal: p-value for H0: difference = 0

ROC

|              | LDA     | ElasticNet | RandomForest | GBM        |
|--------------|---------|------------|--------------|------------|
| LDA          |         | -3.983e-05 | -1.914e-03   | -2.538e-03 |
| ElasticNet   | 1.00000 |            | -1.874e-03   | -2.498e-03 |
| RandomForest | 0.07721 | 0.08125    |              | -6.238e-04 |
| GBM          | 0.03812 | 0.04356    | 1.00000      |            |

Sens

|              | LDA      | ElasticNet | RandomForest | GBM       |
|--------------|----------|------------|--------------|-----------|
| LDA          |          | 0.000389   | -0.022775    | -0.008381 |
| ElasticNet   | 1.000000 |            | -0.023164    | -0.008770 |
| RandomForest | 0.001598 | 0.003055   |              | 0.014393  |
| GBM          | 0.006523 | 0.019979   | 0.001068     |           |

Spec

|              | LDA     | ElasticNet | RandomForest | GBM        |
|--------------|---------|------------|--------------|------------|
| LDA          |         | 0.0002122  | 0.0185307    | 0.0056230  |
| ElasticNet   | 1.00000 |            | 0.0183185    | 0.0054108  |
| RandomForest | 0.01421 | 0.01206    |              | -0.0129078 |
| GBM          | 0.20535 | 0.31973    | 0.06785      |            |

After evaluating the summary of differences between models. There are multiple conclusions



that may be drawn. ROC:

LDA vs. GBM : p value is 0.03812. Difference is -2.538e-03. This is statistically significant ( $P < 0.05$ ) and despite that small difference in ROC, GBM does outperform LDA.

Elastic Net vs. GBM: p-value is 0.04356. Difference is -2.498e-03. This difference in overall performance is statistically significant ( $P < 0.05$ ) and GBM does outperform elastic net.

Sensitivity: GBM vs. LDA: p-value is 0.006523. Difference is -0.008381. The result is statistically significant ( $P < 0.05$ ) and despite the small difference, GBM does outperform LDA in sensitivity.

Random Forest vs. LDA: p-value is 0.001598 Difference is -0.022775 The result is statistically significant ( $P < 0.05$ ) and despite the small difference, Random Forest does outperform LDA in sensitivity.

Random Forest vs. Elastic Net: p-value is 0.003055 Difference is -0.023164 The result is statistically significant ( $P < 0.05$ ) and despite the small difference, Random Forest does outperform Elastic Net in sensitivity.

GBM vs. Elastic Net: p-value is 0.019979 Difference is -0.008770 The result is statistically significant ( $P < 0.05$ ) and despite the small difference, GBM does outperform Elastic Net in sensitivity.

GBM vs. Random Forest: p-value is 0.001068 Difference is -0.0129078 The result is statistically significant ( $P < 0.05$ ) and despite the small difference, GBM does outperform Random Forest in sensitivity.

Specificity: Random Forest vs. LDA: p-value is 0.01421 . Difference is 0.0185307. The result is statistically significant ( $P < 0.05$ ) and despite the small difference, LDA does outperform Random Forest in specificity

Random Forest vs. Elastic Net: p-value is 0.01206 . Difference is 0.0183185 The result is statistically significant ( $P < 0.05$ ) and despite the small difference, Elastic Net does outperform Random Forest in specificity

Exporting the models to model\_evaluation

```
# Set working directory to ensure it saves to a project data folder
setwd(r"(C:\Users\User\ADS503_Team5\Raw Data)")
# Save evaluation file
save(
  mod_glm, mod_glmnet, mod_rf, mod_gbm, models,
  bestTunes, calibration_df, test_y, resamples_diff, res, roc_list,
  file = "evaluated_models.RData"
)
```